# Live Class

# Divisibility Rule ( All PYQ )

## Placement & Competitive Exams

(Infosys | Wipro | TCS | Cognizant | IBPS – IT Officer | Other Placement & Govt Exams)



Campusmonk

# Updated Content: Number System

# Divisibility Rule ( 2 , 4 ,8 )

# Divisibility Rule ( 3 & 9 )

# Divisibility Rule ( 11 )

# Co-prime & Composite Number's

***Question 1:***

What least value must be assigned to * so that the number **451*603** is exactly divisible by 9?

(a) 7 (b) 8 (c) 5 (d) 9

***Question 2:***

If **5432*7** is divisible by 9, then the digit is place of * is ?

(a) 0 (b) 1 (c)6 (d) 9

***Question 3:***

Find the value of k in the number

9314k8025 so that the number is divisible

by 11

***Question 4:***

If the nine-digit number **259876P05** is completely divisible by 11, then what is the value of $(p^2 + 5)$?

A. 58

B. 56

C. 76

D. 54

## Question 5:

**Which of the given values Is divisible by 15?**

A. 2365

B. 1375

C. 4365

D. 2275

## *Question 6:*

**Which of the given values Is NOT divisible by 18?**

**A. 54036**

**B. 50436**

**C. 34056**

**D. 65043**

***Question 7:***

**Which of the following number is divisible by 24 ?**

**A. 35718**

**B. 63810**

**C. 537804**

**D. 3125736**

# Concept Clarity Sheet- 7 , 13

10920

6489

380247

***Question 8:***

**The least number of 5 digits which exactly divided by 52, 56, 78 and 91?**

A. 10290
B. 10860
C. 10920
D. 10580

***Question 9:***

**The Largest number of four digits exactly divisible by 88 is ?**

**A. 9988**

**B. 9944**

**C. 8888**

**D. 9768**

## ***Question 10:***

**If a 10-digit number 5432y1749x is divisible by 72, then what is the value of (5x-4y)?**

**Question 11 :**

**If the number 6p5157q is divisible by 88,**

**then p x q= ____,**

**where p and q are single digit numbers.**

A. 16

B. 18

C. 28

D. 64

E. None of these

**Question 12 :**

**If a 9-digit number 389x6378y is divisible by 72, then the value of 6x + 7y is:**

A. 16

B. 32

C. 28

D. 64

E. None of these

## Question 13 :

If **3236A187B** is divisible by **144,** find the values of A and B respectively.

(a) (2, 4)

(c) (4, 4)

(b) (4, 2)

(d) None of these

***Question 14:***

A number **N = 897324P64Q** is divisible by both 8 and 9. Which of the following is the value of P + Q?

(i). 2 (ii). 11 (iii). 9

(a) either (i) or (ii)

(b) either (ii) or (iii)

(c) either (i) or (ii) or (iii)

(d) None of these

***Question 15:***

**How many pairs (A, B) are possible in the number 89765A4B**

**if the number is divisible by 9 and it is given that the last digit of the number is even?**

**Ans:**

***Question 16:***

When 335 is added to 5A7 the result is 8B2.

8B2 is divisible by 3.

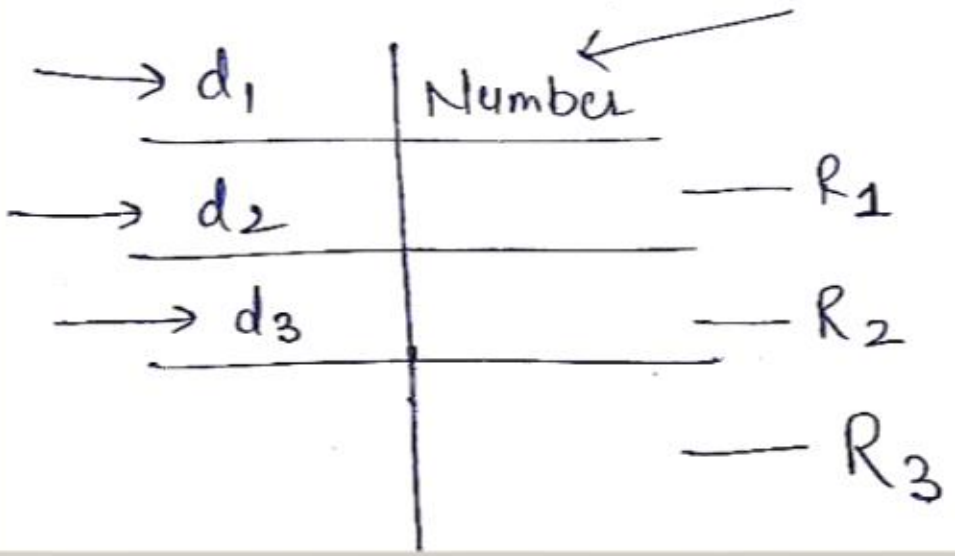
What is the Largest and the smallest possible value of A?

***Question 17:***

$(2^{71} + 2^{72} + 2^{73} + 2^{74})$ is divisible by

(a) 9 (b) 10 (c) 11 (d) 13

## Successive Division :-



Question:

Find the remainder when 2479 is divided successfully by 3, 5, 7 and 2, ?

A. (1, 1, 4, 1)

B. (1, 2, 4, 1)

C. (2, 1, 3, 1)

D. (1, 1, 3, 0)

Question: A number being successively divided by 3, 5 and 6 leaves remainders 1, 3 and 2 respectively. Find the respective remainders if the order of divisors be reversed ?

**A. (4, 1, 1)**
**B. (1, 4, 1)**
**C. (4, 2, 1)**
**D. (2, 1, 4)**

**Question:**

A number when divided successively by 4 and 5 leaves remainders 1 and 4 respectively. When it is successively divided by 5 and 4, then the respective remainders will be?

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***Question 23:***

**Question from UPSC Prelims 2023 CSAT**

**For any choices of values of X, Y and Z, the 6-digit number of the form XYZXYZ is divisible by :**

A. 7 and 11 only

B. 11 and 13 only

C. 7 and 13 only

D. 7, 11 and 13

**Question 24 :**

The number 823p2q is exactly divisible by **7, 11 and 13**. What is the value of (p-q)?

A. 8

B. 5

C. 3

D. 11

***Question 25:***

If the 6 - digit number 57zxy8 is divisible by each of 7,11 and 13, then (x - 2y + z) is:

(a) -1

(c) 1

(b) 2

(d) -2

# Consecutive Number:

1. Average of 9 consecutive positive integer is 63. The product of largest and smallest integer is?

(a) 8935 (b) 3953

(c) 3853 (d) 3845

2. Average of 4 consecutive even No's P, Q, R, S (increasing order) is 51. What is product of P and S?

(a) 2592 (b) 2594

(c) 5293 (d) 2596

3. The average of 37 consecutive No is 54. The largest of these number is?

(a) 72 (b) 70

(c) 74 (d) 76

4. Average of 41 consecutive odd No. is 49. Which of the following is the largest?

(a) 87 (b) 85

(c) 83 (d) 89

5. Average of 44 consecutive odd No. is 144. What is the largest?

(a) 189 (b) 185

(c) 187 (d) 188

6. Average of 14 consecutive even No's is 107. Then which of the following is the smallest Number?

(a) 92 (b) 90

(c) 94 (d) 96

7. Average of 38 consecutive even No's is 167. Find the sum of smallest and largest Numbers?

(a) 167 (b) 334

(c) 332 (d) 336

8. Average of 29 consecutive even No. is 446. Find the difference between first and last Number?

(a) 56 (b) 58

(c) 60 (d) 54